Structure-informed CRISPR co-dependency maps the ARID1A-associated BAF module across cancer cell lines

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Abstract

ARID1A is recurrently inactivated in non-small cell lung cancer, but the functional organization of the ARID1A-associated BAF complex is incompletely resolved. We applied five methodologically distinct analyses to genome-scale CRISPR dependency profiles. All converged on positive co-dependency between ARID1A and core-shared or cBAF-associated subunits, whereas PBAF- and ncBAF-specific subunits were null or negative. The pattern was robust across 19 tissues, two processing resources, and explicit pan-essentiality and lineage controls. In the assembled human BAF structure, ARID1A-contacting subunits had larger co-dependency correlations than non-contacting or mutually exclusive subunits (lung median difference 0.436, 95% bootstrap interval 0.293–0.584; Cliff's $\delta = 0.86$, exact $P = 0.0037$). Contact count did not show a positive dose–response among the six contacts resolved in the same structure, defining structural contact as a binary prior rather than a quantitative predictor. Restricting the core module to load-bearing members improved its anchor-excluded correlation, and the identity of a dilutive member was more reproducible for 227 curated complexes than for size-matched random sets. Genotype-conditional tests did not survive multiple-testing correction; the results therefore map co-essential organization and do not establish an ARID1A-loss synthetic-lethal vulnerability. This distinction provides a rigorous framework for using CRISPR co-dependency to refine protein-complex modules.

Keywords: ARID1A; SWI/SNF (BAF) complex; non-small cell lung cancer; co-dependency; co-essentiality; protein complex; chromatin remodeller

1 Introduction

The SWI/SNF (BAF) chromatin-remodelling complex carries an inactivating mutation in at least one subunit in roughly 20% of human cancers [1], and is now an actively pursued therapeutic target class [2, 3]. It assembles modularly into three mutually exclusive forms — canonical BAF (cBAF), polybromo-associated BAF (PBAF) and non-canonical BAF (ncBAF) — sharing a set of core subunits and each carrying defining subunits [4–6]. These defining subunits are biologically

mutually exclusive: a single complex never contains both ARID1A and PBRM1. This exclusivity is the interpretive basis for the structural analysis below.

ARID1A defines cBAF and is recurrently inactivated in NSCLC [7], where SWI/SNF alterations are associated with poorer immunotherapy outcomes [8]. The lesion itself is not directly druggable, and genetic-interaction studies have therefore sought dependencies associated with the ARID1A-mutant state [9]. A separate and more general question is how ARID1A dependency covaries with other subunits of the assembled BAF complex across heterogeneous cancer cell lines. This continuous co-dependency describes co-essential organization; it is not equivalent to a genotype-conditional synthetic-lethal effect.

Two paradigms motivate different genetic tests. The paralogue synthetic-lethality paradigm holds that ARID1A-mutant tumours depend on ARID1B [10], a relationship since generalised into systematic paralogue-dependency maps [11, 12]. The residual-complex compensation paradigm holds instead that cBAF loss drives compensatory upregulation of ncBAF/PBAF and a switch in dependency to those residual complexes. The two point to different target sets, but loss-conditioned dependency and continuous co-dependency must be analysed and interpreted separately.

Any study of complex-specific dependency also confronts three standing objections: **circularity** (pre-specifying a subunit list makes "discovering" a complex dependency vacuous), the **pan-essentiality artefact** (essential genes covary for reasons unrelated to shared membership), and **lineage confounding** (cell lines cluster by tissue, and module dependencies vary by tissue) [13–15].

A fourth problem receives almost no attention. Module-based analysis requires a *definition* of the module, inherited from biochemistry and used as a fixed input [16]. Nobody asks whether it is the right definition for a dependency analysis: a subunit may be a bona fide member by co-purification yet contribute nothing to — or actively dilute — the dependency signal, because it does not contact the anchor or because paralogues cover its loss.

We address all four. Without pre-specifying any gene list, five methodologically distinct genome-scale analyses are required to converge; pan-essentiality and lineage controls are built in; and the curated module definition is tested rather than assumed. These analyses share a dependency matrix and are therefore convergent views of one functional-genomic measurement, not five independent biological experiments. The experimental structure supplies an orthogonal test and nominates candidate members for refinement; we show that its second role is heuristic rather than decisive. Figure 1 sets out the design and the control applied at each stage.

2 Results

2.1 ARID1A is recurrently mutated and frequently truncated in NSCLC

In the combined TCGA-LUAD and TCGA-LUSC cohort ($n = 999$) [17], *ARID1A* was mutated in **6.51% of cases (65/999)**, at closely similar frequencies in the two histologies (LUAD 6.23%, 32/514; LUSC 6.80%, 33/485; Figure 2a), establishing ARID1A as a recurrently altered gene in this disease context. The rate replicates outside TCGA: two independent lung adenocarcinoma cohorts queried through the cBioPortal API [18, 19] give 5.38% (10/186, MSK-IMPACT [20]) and

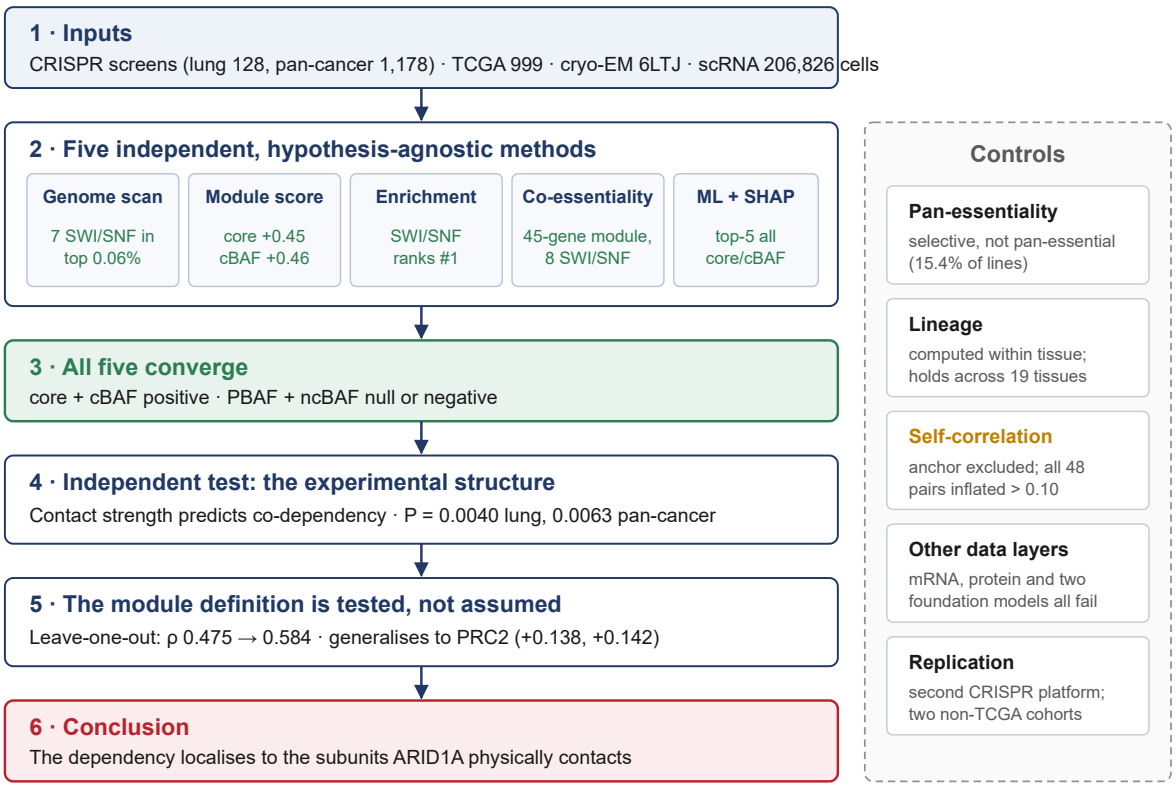


Figure 1: Study design. Five methodologically distinct, hypothesis-agnostic analyses of the same genome-scale dependency matrix are required to converge (stages 1–3); an experimentally determined structure, measured independently of the genetics, then tests that conclusion and is used to interrogate the curated module definition (stages 4–5). The right-hand rail lists the controls applied throughout: no gene list is pre-specified at the discovery stages, which keeps the complex-level conclusion non-circular.

6.67% (177/2,653 [21]) — both within a point of the TCGA estimate despite different platforms, populations and ascertainment.

The mutational spectrum is that of a tumour suppressor (Figure 2b): of 73 variants, **54.8% (40/73) were truncating**, distributed without hotspots, against ∼68% for *STK11* in the same cohort and almost entirely hotspot missense for *KRAS*. The spectrum is consistent with frequent loss of function, although mutation status alone does not establish complete loss of ARID1A protein in an individual model.

Stratifying TCGA-LUAD by *ARID1A* expression tertile (low $n = 180$ versus high $n = 180$), DESeq2 [22] detected **501 significant genes** (adjusted $P < 0.05$, $|\log_2 \mathrm{FC}| > 1$; 421 up, 80 down; Figure 2c). This associative contrast establishes a transcriptional context for low ARID1A expression but is not interpreted as a causal perturbation.

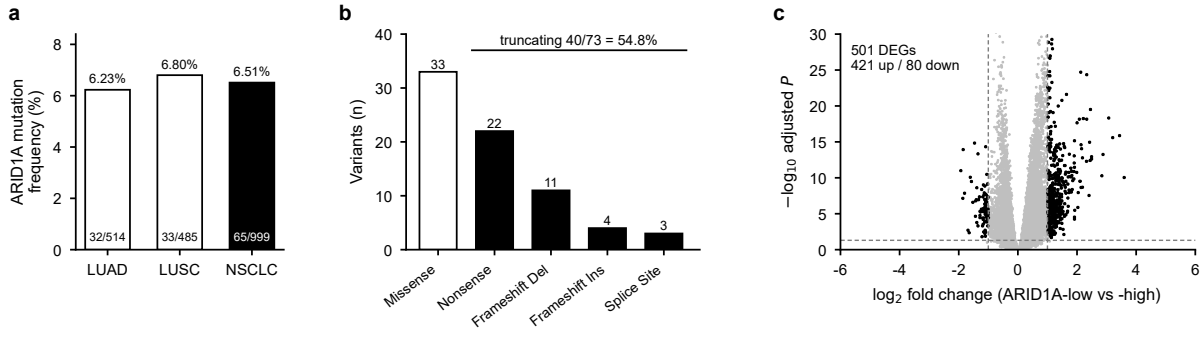


Figure 2: ARID1A is recurrently mutated and frequently truncated in NSCLC. (**a**) Mutation frequency in TCGA-LUAD, TCGA-LUSC and the combined NSCLC cohort; counts give mutated cases over cohort size. (**b**) Variant classification spectrum; filled bars are truncating classes. (**c**) Volcano plot of ARID1A-low versus ARID1A-high tertile differential expression; filled points pass adjusted $P < 0.05$ and $|\log_2 \mathrm{FC}| > 1$.

2.2 Five methodologically distinct analyses converge on the ARID1A-associated BAF module

Five analyses were run separately on the same dependency matrix; four of the five use no complex composition at all (Figure 3). Supplementary Figure S1 gives the sub-complex compositions and the structure of the convergence argument.

Genome-scale scan. Correlating the ARID1A dependency profile from the Project Score screen [23, 24] against ∼17,644 genes in lung cancer ($n = 128$), the top-ranked genes were SMARCE1 (#1, $\rho = 0.551$, $P = 1.6 \times 10^{-11}$), SMARCB1 (#2, 0.474), SMARCC1 (#3, 0.380), DPF2 (#4, 0.363), SMARCA4 (#5, 0.357), ARID1B (#7, 0.344) and SS18 (#10, 0.313) — seven SWI/SNF subunits inside the top 0.06% of the genome (hypergeometric enrichment in the top 1%: $P = 3.2 \times 10^{-12}$ lung, 5.5×10^{-14} pan-cancer), while PBAF- and ncBAF-specific subunits ranked in the bottom 90–97%. The pan-cancer ranking has the same shape (Supplementary Figure S2): eight of the top nine genes are SWI/SNF subunits, with the paralogue ARID1B only 32nd. Among the co-dependencies for which a chemical probe or clinical-stage inhibitor exists, SMARCA4 ranks first; we list these as starting points for combination work, not as findings of this study.

Module scoring. Core-shared and cBAF-specific modules were significantly positive at both levels; PBAF-specific and alternative-ATPase modules null; the ncBAF-specific module significantly negative pan-cancer (Table 1).

Table 1: Complex-module co-dependency with ARID1A. The anchor is excluded from every module (Section 2.7).

Module	Subunits	Lung ρ	P	Pan-cancer ρ	P
Core-shared	8	+0.447	1.2×10^{-7}	+0.461	3.1×10^{-50}
cBAF-specific	9	+0.456	6.2×10^{-8}	+0.307	1.0×10^{-21}
PBAF-specific	4	−0.123	0.166	−0.012	0.720
ncBAF-specific	3	−0.157	0.076	−0.110	8.1×10^{-4}
Alt. ATPase (SMARCA2)	1	−0.122	0.170	−0.002	0.958

Prior-free enrichment. The top-100 co-dependent genes, submitted to Enrichr [25] without context, returned SWI/SNF Complex (GO:0016514) as the top term in lung (9/29 genes, adjusted $P = 1.5 \times 10^{-12}$), followed by nBAF, CORUM EBAFb, npBAF and BAF — every leading term a SWI/SNF sub-complex.

Co-essentiality community detection. From 7,940 selectively essential genes we built a genome-wide k-nearest-neighbour co-dependency graph ($K = 15$, $\rho > 0.15$) and ran Leiden community detection [26] across a resolution sweep. At resolution 4.0 the ARID1A community contained **45 genes including 8 SWI/SNF subunits** (hypergeometric $P = 1.06 \times 10^{-13}$). The algorithm is given neither the anchor's identity nor the study question.

Gradient boosting with SHAP. Predicting ARID1A dependency from 3,000 genome-scale features (1,178 lines) gave cross-validated $R^2 = 0.376$. The top five SHAP features [27] were SMARCB1, SMARCE1, SMARCC1, ARID1B and DPF2, separated by a clear discontinuity from rank 6. PBAF/ncBAF subunits ranked far lower (BRD9 #30, ARID2 #108, BRD7 #224, PBRM1 #301). Because the original training driver was not retained, we performed a fully specified pre-submission reanalysis with unsupervised variance-based feature selection and out-of-fold prediction. It recovered the same five leading features and a similar out-of-fold $R^2 = 0.390$, providing a reproducible sensitivity analysis rather than relying on the historical model output alone.

Mandatory controls. ARID1A is *selectively*, not pan-, essential — dependent (gene effect < -0.5) in 15.4% of lines and strongly dependent (< -1.0) in 1.3% — whereas pan-essential references are essential in 100% of lines, so its correlations cannot be a pan-essentiality artefact. Correspondingly ARID1A co-dependency is strong with core BAF and null with every pan-essential reference ($|\rho| \leq 0.070$) and with PBRM1 (+0.002). Lung estimates are computed within lineage; robustness to broader lineage composition is assessed by separate within-tissue analyses across 19 tissues below.

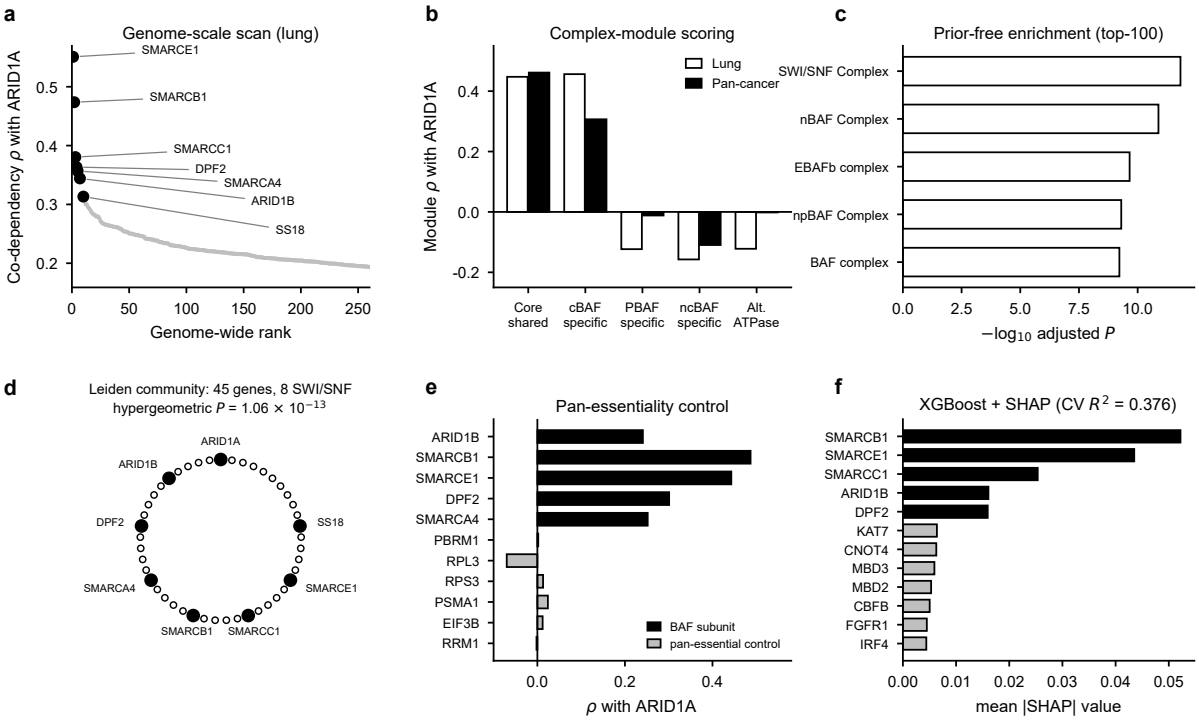


Figure 3: Five methodologically distinct analyses converge. (**a**) Genome-scale co-dependency scan in lung cancer; SWI/SNF subunits are highlighted. (**b**) Complex-module scores in lung and pan-cancer. (**c**) Prior-free enrichment of the top-100 co-dependent genes. (**d**) Leiden community containing ARID1A. (**e**) Pan-essentiality specificity controls. (**f**) XGBoost/SHAP feature attribution.

2.3 The experimental structure explains the genetic result and refines the module definition

The preceding section rests entirely on genetic correlation, so we introduced evidence of an entirely different kind: the cryo-EM structure of the assembled human BAF complex (PDB 6LTJ, 3.70 Å; base module 6LTH, 3.00 Å) [28], with the nucleosome-bound PBAF structure [29] as a negative control — it contains no ARID1A chain at all, confirming cBAF/PBAF mutual exclusivity structurally rather than by assumption.

Parsing chain coordinates with Biopython [30], ARID1A makes direct heavy-atom contact (minimum distance 2.20–2.47 Å) with SMARCC2 (754 atom pairs within 4.5 Å), SMARCA4 (557), SMARCD1 (553), SMARCB1 (300), SMARCE1 (200) and DPF2 (137), and no contact with ACTL6A (13.02 Å) or ACTB (31.99 Å) (Figure 4I). SMARCC2 is present as a homodimer (chains N and O). Independent recomputation in the higher-resolution 6LTH gave identical counts to within one atom pair. SMARCC1, unresolved in 6LTJ, contacts ARID1A in the integrative model of the endogenous complex (PDB 9A0K; 186 pairs, 2.58 Å) [31]; the two agree on all six subunits 9A0K models to comparable depth, diverging only for SMARCE1 and DPF2, which it covers over 79 and 37 residues. We count SMARCC1 as contacting.

Physical contact separates stronger from weaker co-dependencies. Comparing the seven contacting genes against eight non-contacting or mutually exclusive ones gave a lung median difference in ρ of 0.436 (95% bootstrap interval 0.293–0.584; Cliff's $\delta = 0.857$; exact two-sided Mann–

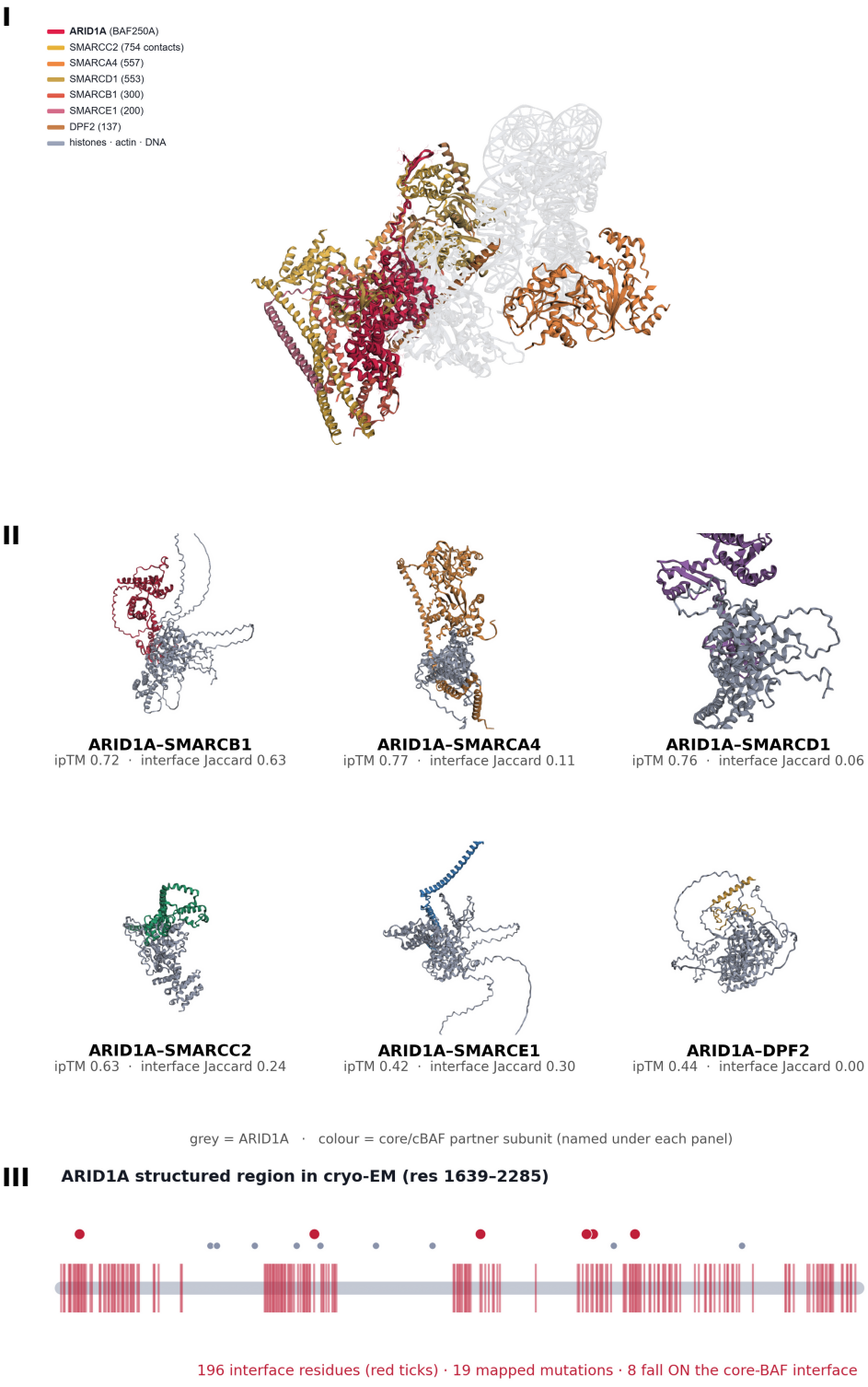


Figure 4: The ARID1A–core-BAF interface: experimental, predicted, and mutated. (**I**) ARID1A within the assembled human cBAF complex (PDB 6LTJ, 3.70 Å) [28], ARID1A in red, each contacting subunit coloured and annotated with its heavy-atom contact count (< 4.5 Å); histones, actin and DNA grey. ARID1A contacts all six core/cBAF subunits at 2.20–2.47 Å, while ACTL6A and ACTB lie 13.0 and 32.0 Å away. (**II**) The same six complexes predicted by AlphaFold-Multimer [32–34] from sequence alone, with self-reported ipTM and Jaccard overlap against the cryo-EM interface. The two highest-confidence models reproduce it worst; the best reproduction ranks only third by ipTM. (**III**) The ARID1A region resolved in 6LTJ (residues 1639–2285): red ticks are the 196 interface residues, circles the 19 NSCLC mutations in this region, eight of them (filled) on the interface.

Whitney $P = 0.0037$) and a pan-cancer difference of 0.314 (0.165–0.451; $\delta = 0.821$; $P = 0.0059$). Restricted to the six subunits resolved in 6LTJ, the group comparisons gave $P = 0.0080$ and $P = 0.0127$. This binary structural partition is an orthogonal measurement to the genetic screen. It should not be interpreted as a dose–response: among the six contacting subunits resolved in the same structure, contact count was not positively associated with co-dependency (lung Spearman $\rho = -0.657$, $P = 0.156$; pan-cancer $\rho = -0.600$, $P = 0.208$). Physical contact is therefore used as a binary prior, not as a quantitative predictor of dependency strength.

Curated module definitions are not optimal. Restricting the core module to the subunits the structure shows to contact the anchor gives a better module: on DepMap (lung $n = 126$; pan-cancer $n = 1{,}178$) the correlation rises from 0.475 to 0.584 (lung) and 0.499 to 0.533 (pan-cancer) while dropping two of eight genes (Figure 5c). ACTB and ACTL6A are individually null (lung $\rho = 0.043$ and 0.122) and act as dilution terms: in lung, leave-one-out over those same eight (Figure 5b) gains +0.061 and +0.037 on their removal while removing a contacting subunit costs up to −0.089. The separation is cleaner in lung than pan-cancer, where ACTL6A costs −0.007. SMARCD1 is an apparent exception — despite the second-largest interface its removal improves the pan-cancer score, matching its null co-dependency ($\rho = -0.083$ lung, −0.025 pan-cancer). Separating the paralogues resolves it: the dependency at that position runs through SMARCD2, which no cBAF structure resolves ($\rho = +0.193$, $P = 2.9 \times 10^{-9}$ pan-cancer, tenth of 17,644 genes; +0.184, $P = 0.038$ in the smaller lung panel). The structure localises the position, not the gene occupying it.

The refinement generalises to a second complex. Repeating the procedure on PRC2 with the cryo-EM structure of the human PRC2–AEBP2–JARID2 complex [35, 36] and two anchors (Figure 5d), both converge on the catalytic lobe EZH2–EED–SUZ12, with the histone-chaperone subunits RBBP4 and RBBP7 removed: EZH2 improves from 0.607 to 0.746 and SUZ12 from 0.581 to 0.723. RBBP4/RBBP7 are shared with NuRD, CAF-1 and SIN3, so their dependency profiles are driven by complexes other than PRC2.

How far does structural contact go as a decision rule? For two of the three anchors it is exactly right: for ARID1A the contact-restricted core is the best-performing definition, and for EZH2 the contact rule selects EED + SUZ12, which an exhaustive subset search confirms is the optimum. For the third it is not: RBBP4 makes a large, unambiguous contact with SUZ12 (685 heavy-atom pairs across 61 interface residues) and would be retained, yet removing it improves the module by a further +0.092. Membership in other complexes is the obvious explanation, and it does not survive testing (Discussion).

Operational recommendation. Structural contact is a good but imperfect prior — correct for two of three anchors — and the criterion should be whether the dropped member replicates on resampling, not the size of the gain, for the reason given in the Discussion. The refinement is substantial and consistent: +0.109 (SWI/SNF, lung), +0.138 (PRC2, EZH2) and +0.142 (PRC2, SUZ12) over the curated definition.

The reproducibility criterion generalises beyond the two examples. Across 227 curated human complexes with sufficient DepMap coverage, choosing the largest leave-one-out gain was not informative by itself: size-matched random gene sets gained at least as much (median +0.026 versus +0.019). In contrast, the identity of the member nominated as dilutive replicated across random half-splits with median concordance 0.70, compared with 0.30 for random sets and 0.125 expected from module size ($P = 9.8 \times 10^{-13}$). Thus the transferable result is a resampling criterion for module refinement, not a universal rule that non-contacting members should be removed.

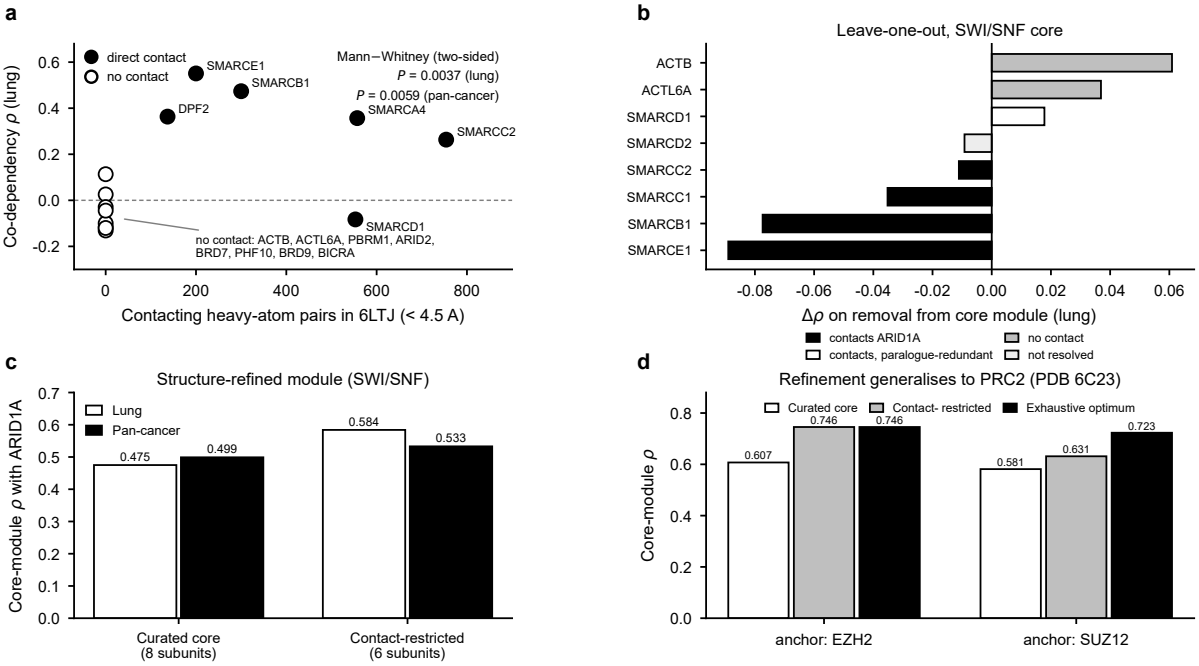


Figure 5: The experimental structure explains and refines the genetic result. (**a**) Contacting heavy-atom pairs in PDB 6LTJ against co-dependency ρ; filled points contact ARID1A directly. The inferential test compares contacting with non-contacting subunits; contact count within the contacting group does not show a positive dose–response. SMARCC1 is not plotted because 6LTJ does not resolve it; its contact is established in 9A0K. (**b**) Leave-one-out change in the core-module correlation in lung, over the same eight subunits used in (**c**), ordered by contact status; SMARCD2 is resolved in no cBAF structure and so has no contact call. (**c**) Curated versus contact-restricted core module in SWI/SNF. (**d**) The same refinement in PRC2 (PDB 6C23) for two anchors, against the exhaustive subset optimum.

2.4 The continuous co-dependency pattern varies with ARID1A expression

If co-dependency reflects shared complex state, its magnitude may vary with the abundance of the anchor. Across *ARID1A* expression quintiles the core-module dependency changes strictly monotonically (-0.706 to -0.510 from lowest to highest), as does the cBAF module, while PBAF (≈ -0.19) and ncBAF ($\approx +0.01$) are flat. The trend is confined to the modules implicated by the co-dependency analysis; partial correlation preserves the specificity (pan-cancer core 0.411, $P = 3.6 \times 10^{-49}$; cBAF 0.218, $P = 4.1 \times 10^{-14}$). Because expression is observational and does not identify protein-null models, this gradient is a consistency analysis rather than evidence of loss-induced sensitivity.

2.5 Generalisation across tissues and independent replication

Across 930 de-duplicated lines from 19 tissues, computed within each tissue, SMARCB1 and SMARCC1 were positive in 19/19 tissues, DPF2 in 18/19, ARID1B 17/19 and SMARCA4 16/19, three to thirteen of them passing global Benjamini–Hochberg correction [37]; PBRM1, ARID2 and SMARCA2 were each positive in 10/19 and negative in 9/19 with *zero* significant tissues (Figure 6a).

Recomputation in a wholly different dataset and pipeline — the DepMap 24Q4 Chronos gene-effect matrix, 1,178 lines [38] — reproduced the pattern (Figure 6b): core/cBAF positive, PBAF/ncBAF/alternative ATPase null, so the finding is not an artefact of one processing pipeline.

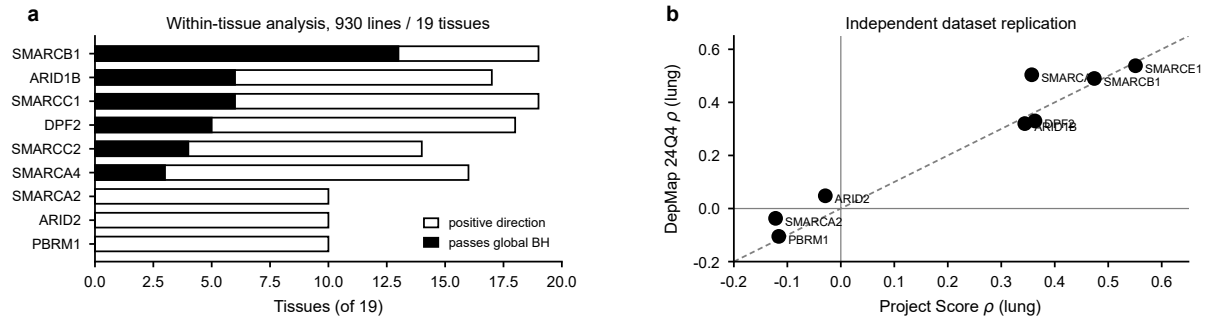


Figure 6: Pan-cancer generalisation and independent replication. (**a**) Within-tissue co-dependency across 19 tissues; open bars count tissues with a positive direction, filled bars those passing global BH correction. (**b**) Project Score versus DepMap 24Q4 co-dependency in lung; the dashed line is the identity.

2.6 Genotype-conditional tests bound the co-dependency interpretation

We next asked the distinct question of whether ARID1A-mutant cell lines are more dependent on individual BAF subunits than wild-type lines. Across five tissues with at least six mutant models, ARID1B had the expected direction in all five, ARID2 and BRD9 in four, and the remaining subunits in one to three. None of the 65 tissue–subunit comparisons survived global Benjamini–Hochberg correction. In lung ($n = 6$ mutant, $n = 122$ wild type), PBRM1 and ARID2 were nominally more essential in mutant lines ($P = 0.0217$ and 0.0248), but both had $q = 0.403$; the principal contacting subunits were non-significant. Thus the continuous co-dependency pattern is robust, whereas this dataset does not establish an ARID1A-loss-conditioned vulnerability. Mutation status also combines variants with different consequences for protein abundance, making protein-defined or isogenic loss models the appropriate next test.

2.7 A self-correlation artefact must be corrected

Module scoring correlates an anchor's dependency against a module mean. If the anchor is itself a member, $1/k$ of the module score *is* the anchor. Across three complexes and 48 anchor–module pairs all 48 were inflated by more than 0.10, with inflation scaling inversely with module size (Spearman $\rho = -0.597$, $P = 7.3 \times 10^{-6}$; mean +0.506 at $k = 2$, +0.358 at $k = 3$, +0.197 at $k = 9$). At small k the consequence is a reversed conclusion, not a shifted estimate: STAG2 appears strongly co-dependent with its own cohesin module (+0.839) when the anchor-excluded

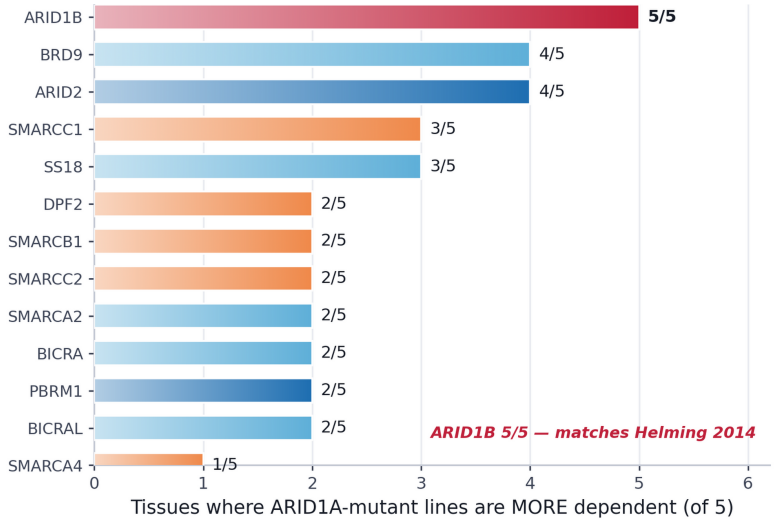


Figure 7: Genotype-conditional dependency is directional but underpowered. For each BAF subunit, the number of tissues (of five with adequate sample size) in which ARID1A-mutant cell lines are more dependent than wild-type lines. ARID1B has the expected direction in 5/5 tissues, but no tissue–subunit comparison survives global multiple-testing correction. This analysis measures a different axis from the continuous co-dependency gradient and is reported as a boundary on interpretation, not as evidence of synthetic lethality.

value is -0.045, not significantly different from zero. All analyses here exclude the anchor from every module.

That correlating a variable with a mean containing it inflates the correlation is elementary; the contribution here is the empirical magnitude and the demonstration that at small k it reverses conclusions. We have not surveyed how often the anchor is left in, and make no claim about prevalence.

Benchmark with the correction applied. The shared core module is common to all sub-complexes, so a high core score is expected for *every* anchor; the testable hypothesis is one of discrimination. Scored over fourteen anchors of known mutually exclusive membership across SWI/SNF and PRC2, all fourteen are assigned correctly, competing modules near zero or negative throughout, and SMARCA2 returns null on all four modules ($|\rho| \leq 0.042$) — correct for an alternative ATPase belonging to no sub-complex. Since co-essentiality recovering complex membership is established [13, 14], this is a positive control, not a novel finding.

Null controls. Three hundred random genes outside the complex give a best exclusive-module correlation of $+0.028$ (median; 95th percentile $+0.095$); only 4.0% pass a $\rho > 0.1$ threshold and only 0.3% exceed the weakest true anchor (Mann–Whitney $P = 1.7 \times 10^{-7}$). Decoy anchors from unrelated complexes give a median of $+0.091$, but not all are inert: MED12 scores $+0.241$ on ncBAF and the three PRC2 core subunits lean toward ncBAF ($+0.138$ to $+0.181$). Thresholds in the 0.1–0.25 range therefore admit false positives from functionally related complexes, and assignments should carry their margin over the best competitor.

2.8 Computational structural methods cross-validate one interface, and one confidence metric fails

Blind rigid-body docking [39] with 200 swarms (6,000 poses per partner) located the correct interface for all four partners tested (best Jaccard 0.325–0.500; Figure 8a). Top-ranked poses perform poorly because the scoring function ranks large scaffold interfaces badly, so we treat this only as evidence that the interface is findable.

AlphaFold-Multimer [32–34] run from sequence alone accurately recovered the ARID1A–SMARCB1 interface (87% recall, Jaccard 0.628). But across the six partners **ipTM and interface recovery are unrelated** (Spearman $\rho = -0.14$, $P = 0.79$; Figure 8b), and the two highest-confidence predictions are among the worst reproductions, consistent with benchmarking showing that multimer confidence metrics degrade for large and asymmetric assemblies [40, 41] and with the incomplete structural coverage of the human interactome [42]. ipTM should not be used as a reliability filter when validating interfaces within large assemblies. The six models are shown in Figure 4II.

The interface is also where a substantial share of the mutations fall: of the 73 ARID1A mutations in the NSCLC cohort, 19 lie within the region resolved in 6LTJ (residues 1639–2285) and eight of those land on the 196-residue core-BAF contact surface (Figure 4III). Because most ARID1A mutations are truncating (Figure 2b) and remove the region entirely, this is not evidence for a separable interface-disrupting mutation class; it records only that the contact surface lies within the mutated territory.

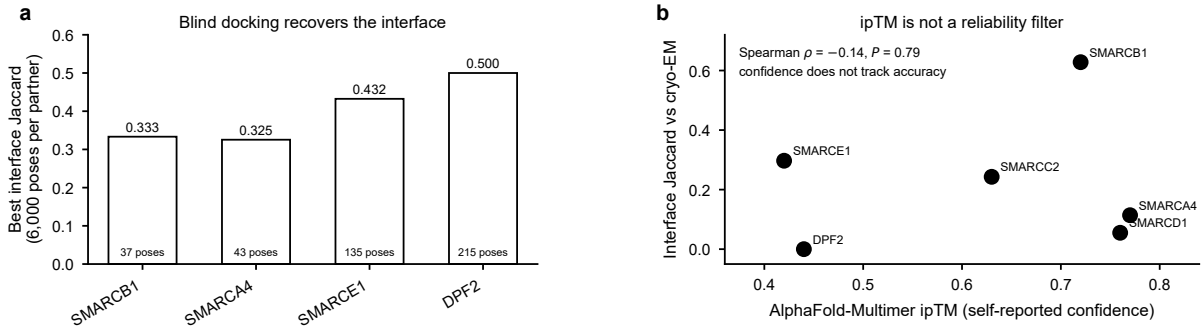


Figure 8: Computational structural cross-validation. (**a**) Blind docking interface recovery; the best Jaccard over all 6,000 poses per partner, with the number of near-native poses ($J \geq 0.2$) annotated. (**b**) AlphaFold-Multimer interface reproduction against self-reported ipTM for the six contacting partners.

2.9 Alternative explanations excluded

The complex the dependency targets is expressed in real tumours. In 206,826 Harmony-integrated cells from a NSCLC single-cell atlas [43, 44], ARID1A and the residual core/cBAF subunits are expressed in the malignant epithelial compartment (Supplementary Figure S3). To respect the nesting of cells within specimens, we aggregated the 35,600 epithelial cells to 48 sample-level observations. Mean ARID1A expression was modestly associated with the mean core/cBAF score (Spearman $\rho = 0.416$, $P = 0.0033$); samples below versus above the median differed by only 0.0147 score units (medians -0.0179 versus -0.0032; exact Mann–Whitney $P = 0.0257$). The small

sample-level effect shows that the genes are detectable in tumour epithelium but does not establish that low ARID1A expression is functional loss or that an intact residual complex persists. Human Protein Atlas immunohistochemistry agrees at the protein level, reporting moderate-to-strong nuclear ARID1A and strong nuclear SMARCB1 and SMARCA4 staining across malignant tissue including lung [45].

Transcript-level data cannot resolve complex specificity. *In silico* deletion of ARID1A with a single-cell foundation model [46] in 7,270 malignant epithelial cells [43] returned null for BAF subunit ranks ($P = 0.3$–0.6) in two independent runs; the undownsampled run increased detection counts three- to four-fold and remained null. An architecturally different foundation model agrees from the opposite direction: in scGPT's representation [47] the SWI/SNF subunits sit close to ARID1A regardless of sub-complex, with PBRM1 its nearest SWI/SNF neighbour despite a genetic PBAF co-dependency of essentially zero (Supplementary Figure S3). Bulk mRNA and cell-line proteomic correlations [48] likewise place ARID1A in positive correlation with *all thirteen* BAF family genes, including the mutually exclusive PBRM1 and SMARCA2 (Figure 9a), so a conclusion about complex specificity drawn from the sign of an expression correlation would be demonstrably wrong at the genetic level.

The specificity is present at the protein layer but not the transcript layer. Sign is not the informative feature; magnitude is. On the 677 cell lines carrying both measurements, the same lines used for both, whether ARID1A contacts a subunit is invisible in the transcriptome (median ρ 0.571 contacting versus 0.432 not; AUC 0.57, $P = 0.69$) and plainly visible in the proteome (0.598 versus 0.207; AUC 0.96, $P = 0.0012$), where the correlation also tracks interface size ($\rho = +0.65$, $P = 0.0086$, against $+0.18$, $P = 0.53$ for transcripts). The comparison is confounded: protein correlation tracks detection rate and abundance ($\rho = +0.82$ for both) and contacting subunits are the better-measured ones. Restricted to the nine subunits detected in over 90% of lines the protein-level difference persists (0.605 versus 0.385, $P = 0.048$) while the transcript-level difference stays absent and reverses in sign, so we report the contrast as supported but abundance-limited.

An exploratory Mendelian-randomization analysis did not provide evidence for a germline ARID1A-expression effect on lung-cancer risk (Figure 9b).

2.10 Pharmacological data do not establish genotype selectivity

Thirteen GDSC compounds targeting PARP, ATR, ATM, EZH2 or SWI/SNF were compared between ARID1A-mutant and wild-type models in lung and pan-cancer strata. No comparison survived within-stratum Benjamini–Hochberg correction ($q \geq 0.27$). PFI-3 and I-BRD9 were among the least active compounds in the lung panel (mean AUC 0.977 and 0.956), leaving little dynamic range for a genotype interaction. These data are therefore uninformative rather than evidence against selectivity. Likewise, genotype-conditional CRISPR effects for SMARCA4 and SMARCA2 did not survive correction. The present study consequently makes no therapeutic-efficacy claim: co-dependency prioritizes functional relationships, not drug targets, and requires protein-loss-defined or isogenic validation before pharmacological translation. The docking, repurposing and short single-trajectory simulations from an earlier exploratory analysis were

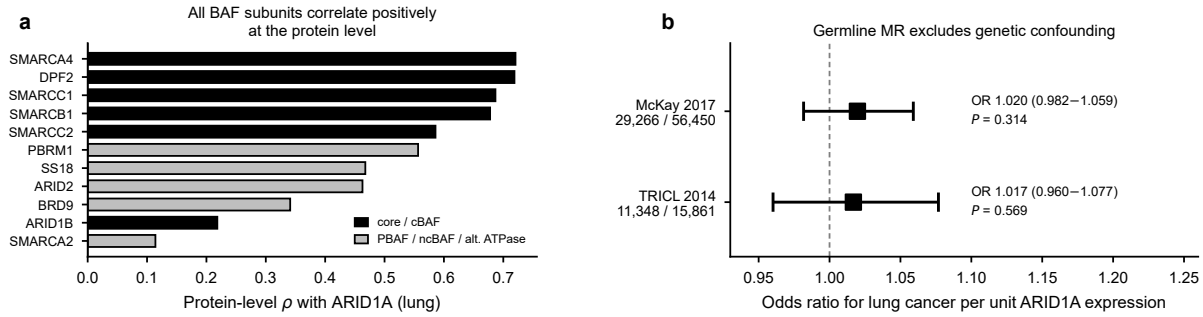


Figure 9: Expression-layer comparison and exploratory germline analysis. (**a**) Protein-level correlation of ARID1A with BAF family subunits in lung cell lines; all subunits correlate positively regardless of sub-complex membership. (**b**) Two-sample Mendelian randomization [49] using ARID1A *cis*-eQTL instruments [50] against two independent lung cancer GWAS [51]; squares are odds ratios with 95% confidence intervals. Both are null (OR = 1.020, $P = 0.314$; OR = 1.017, $P = 0.569$). Only one non-lung instrument was available: ARID1A carries seven independent *cis*-eQTLs across 49 GTEx tissues [52] and none in lung, as expected for a haploinsufficient tumour suppressor under purifying selection [53]. The null estimates are therefore weakly informative and cannot exclude genetic confounding.

removed from the evidentiary chain because they interrogated ATPase/bromodomain pockets rather than the scaffolding relationships identified by the primary analysis.

3 Discussion

Five methodologically distinct analyses of one CRISPR dependency matrix converge on core-shared and cBAF-associated subunits. The pattern survives pan-essentiality and lineage controls, is reproduced in a second processing resource and across 19 tissues, and aligns with a binary contact partition in an experimental structure. Together these results map co-essential organization of the ARID1A-associated BAF module. They do not show that ARID1A loss creates the pattern, because the separate genotype-conditional analysis is underpowered and non-significant after correction.

Curated module definitions can be tested empirically. The 227-complex survey shifts the methodological emphasis from maximizing an in-sample correlation to reproducing the identity of a dilutive member under resampling. The former is guaranteed to improve under best-of-k selection even for random sets; the latter discriminates curated complexes from the matched null. This criterion is useful beyond SWI/SNF, while remaining a statistical refinement rather than proof that the nominated member is biologically dispensable.

Structure is a prior, not a rule. We expected structural contact to supply the decision criterion. In SWI/SNF the prior happens to be correct, because the two dilutive subunits are also the two non-contacting ones; PRC2 dissociates the properties, since RBBP4 contacts SUZ12 substantially yet dilutes. Subunit promiscuity is the obvious candidate, and it fails: the number of curated human complexes containing each subunit [54] — an external count, not one derived from the dependency data — does not predict the leave-one-out delta (Spearman $\rho = +0.21$, $P = 0.63$ within SWI/SNF), and dilutive and load-bearing subunits have indistinguishable counts

($P = 0.91$). Curation depth confounds such counts, so this bounds rather than closes the question. We say so explicitly because the purely structural rule would be the more attractive claim and is not what our data support.

Two layers that do not substitute for one another. Every BAF family gene correlates positively with ARID1A in both mRNA and protein, mutually exclusive subunits included, so the sign of an expression correlation carries no information about complex identity. What separates the layers is that the protein-level magnitude reproduces the structural partition while the transcript-level magnitude does not — an orthogonal measurement type, on the same cell lines, recovering what the CRISPR screens give. Transcriptional co-regulation is a poor proxy for complex membership; protein abundance is a usable one, within the limits of what mass spectrometry detects.

Relationship to existing paradigms. ARID1B shows a *positive* continuous co-dependency with ARID1A, superficially opposite to synthetic lethality [10], and is robust across pipelines. In the separate genotype-stratified analysis ARID1B had the expected direction in all five tissues, but no tissue-specific comparison survived multiplicity correction. The result is compatible with the established ARID1A–ARID1B relationship but does not independently reproduce it. More broadly, continuous co-dependency across heterogeneous lines and conditional essentiality after complete loss are different estimands. Protein-defined null models or isogenic perturbations are needed before translating the former into a synthetic-lethal claim.

Limitations of the study

The study is entirely computational and correlative. Positive co-dependency identifies cell lines in which two genes tend to be essential together; it does not establish that loss of one gene creates dependence on the other. The genotype-conditional tests were underpowered and non-significant after correction, and mutation status does not guarantee loss of ARID1A protein. Protein-defined null models, isogenic perturbations and rescue experiments are the necessary next tests of loss-conditioned vulnerability.

The structural contact analysis supplies an orthogonal physical rationale rather than a mechanism. It uses a static wild-type assembly and therefore cannot show the composition or conformation of a residual complex after ARID1A loss. Contact status separates the co-dependency groups, but contact count does not show a positive dose–response among resolved contacts. Co-immunoprecipitation or native-gel assays would be needed to test complex integrity directly.

Module refinement is characterised on two complexes and three anchors and surveyed across 227; the survey establishes that the dilutive member replicates, not that dropping it is biologically correct, and the mechanism is unresolved. The real-complex survey selected the highest-correlating anchor before resampling, whereas the random-set null used an arbitrary anchor; a future nested analysis should mirror anchor selection in both arms. Two subunits are placed by an integrative model rather than refined density: SMARCC1, which no experimental cBAF structure resolves, and SMARCD2, whose position is inferred from its paralogue. Computational structural methods

validate only the ARID1A–SMARCB1 interface effectively; the 14/14 discrimination benchmark is a positive control rather than a novel finding, and decoy controls are not fully inert. The five discovery analyses share one dependency matrix and are methodologically distinct rather than biologically independent. The single-cell result is sample-aggregated but remains an expression analysis, and the transcript- and foundation-model negatives are unexplained mechanistically.

Methods

Data sources. TCGA-LUAD ($n = 514$) and TCGA-LUSC ($n = 485$) [17]; the Sanger Project Score CRISPR screen, de-duplicated by model identifier (lung $n = 128$; pan-cancer 930 lines, 19 tissues) [23, 24]; the DepMap 24Q4 Chronos matrix (1,178 lines) [38]; PDB 6LTJ, 6LTH, 6C23 and 9A0K; GEO GSE131907 [43]; OpenGWAS accessions for the eQTL exposure [50] and lung cancer outcomes [51].

Mutation landscape and differential expression. maftools [55] on the clinically matched cohort denominator; other MAF sources give slightly different denominators (6.1–6.8%), so the cohort definition must be stated for reproduction. DESeq2 [22] on TCGA-LUAD by *ARID1A* expression tertile, significance at adjusted $P < 0.05$ and $|\log_2 \mathrm{FC}| > 1$.

Independent mutation-frequency replication. Two non-TCGA lung adenocarcinoma cohorts queried through the cBioPortal REST API [18, 19]: `luad_mskimpact_2021` ($n = 186$) [20] and `luad_mskcc_2023_met_organotropism` ($n = 2{,}653$) [21]. A sample was called mutant if it carried at least one non-synonymous *ARID1A* variant; frequency is mutant over all sequenced samples. Both are targeted panels covering ARID1A, ARID1B, SMARCA4, SMARCB1, SMARCD1, PBRM1 and ARID2 (SMARCE1 and SMARCA2 added in v505), which is why the replication is restricted to mutation frequency.

CRISPR co-dependency. Spearman correlation with global Benjamini–Hochberg correction [37]. **Tissue-level correlations are computed within tissue**; pan-cancer values are pooled, the lineage control being the separate 19-tissue analysis. Module score is the mean gene effect over module members, **with the anchor excluded from every module**. Table 1 uses Project Score; the refinement and leave-one-out analyses use DepMap 24Q4, which is why the curated core's correlation differs between them. Prior-free enrichment via Enrichr [25] on the top-100 co-dependent genes only. Co-essentiality network from 7,940 selectively essential genes with Leiden community detection [26], community identity interpreted only after detection. The archived feature-attribution analysis used XGBoost with SHAP [27]. For reproducible sensitivity analysis, the DepMap 24Q4 matrix was restricted to the 3,000 non-target genes with highest variance, missing values were median-imputed, and prediction was evaluated by five-fold out-of-fold cross-validation (shuffle seed 20260831). XGBoost used 500 trees, maximum depth 3, learning rate 0.03, subsample and column-sample fractions 0.8, and L_2 regularisation 1.0; SHAP values were then computed from a model fitted to all 1,178 lines. Null controls: 300 random non-member genes and 13 decoy anchors from PRC2, cohesin, ribosome, proteasome, RNA polymerase II and Mediator,

scored identically.

Structural analysis. Biopython [30] parsing of PDB 6LTJ and 6LTH (SWI/SNF) and 6C23 (PRC2); minimum heavy-atom distance and count of heavy-atom pairs within 4.5 Å between the anchor chain and every other chain; chain identity verified against PDB COMPND records; exact two-sided Mann–Whitney comparison of contacting versus non-contacting groups, with Cliff's delta and a 20,000-resample percentile bootstrap interval for the median difference (seed 20260831); Spearman correlation of contact count with co-dependency was assessed separately among the six contacts resolved in 6LTJ. Leave-one-out and exhaustive-subset module recomputation. In 6C23 subunits are modelled as disjoint fragments rather than separate copies, so per-gene contacts are summed across fragments and a contact required at least three interface residues. The same criterion placed SMARCC1 using the integrative model 9A0K [31]; built from crosslinking mass spectrometry and electron-microscopy envelopes rather than refined density, it is used only for the binary contact call, and modelled residue coverage was compared subunit by subunit first. Subunit promiscuity was the number of EBI Complex Portal complexes containing that protein [54], tested against the leave-one-out delta by Spearman correlation.

Protein versus transcript layer. Sanger cell-line proteomics [48] and CCLE expression were matched through Cell Model Passports identifiers, giving 677 lines with both. Per subunit, the ARID1A-partner Spearman correlation was computed at the protein and the transcript level on the identical lines, so sample composition cannot generate a difference between layers, and the contacting and non-contacting groups compared within each layer by two-sided Mann–Whitney. Because protein correlation is not independent of how well a protein is measured, detection rate and abundance were tested as confounders and the comparison repeated on subunits detected in over 90% of lines.

Generalisation survey. Every Complex Portal human complex with at least six subunits in the DepMap matrix was tested (227 after keeping the highest-correlating anchor per complex), each scored against its anchor-excluded module as above. Reproducibility of the dilutive member was measured by 20 random half-splits of the cell lines per complex, recording how often both halves nominate the same subunit. The comparison used size-matched random gene sets. Because the real complexes were first reduced to their highest-correlating anchor whereas each random set used an arbitrary anchor, the survey is treated as exploratory and this asymmetry is stated in Limitations.

Interface prediction. AlphaFold2-Multimer v3 via ColabFold [32–34] on the ARID1A domain (UniProt 1639–2285) with partner sequences; interface compared on the ARID1A side at 4.5 Å. Blind rigid-body docking with LightDock [39] used 200 swarms and 6,000 poses per partner as an interface-recovery sensitivity analysis.

Protein-level expression in tumour tissue. Immunohistochemistry annotations for ARID1A, SMARCB1 and SMARCA4 were read from the Human Protein Atlas cancer-pathology pages [45]

— the resource's own qualitative staining calls, used only as corroboration that the complex is present at protein level in malignant tissue.

Single-cell analysis. GSE131907 [43] integrated across samples with Harmony [44]; 206,826 cells after quality control. The core/cBAF module score is the mean scaled expression of the eight shared-core subunits per cell. For inference, epithelial-cell ARID1A expression and module score were averaged within each of 48 samples, then assessed by Spearman correlation and an exact Mann–Whitney comparison after a sample-level median split. Cell-level distributions are visualized descriptively only.

Genotype-stratified drug response. GDSC area-under-curve values for thirteen compounds — PARP (olaparib, talazoparib, niraparib, rucaparib, veliparib), ATR (AZD6738, AZ20, VE-822), ATM (KU-55933, KU-60019), EZH2 (GSK343) and the two SWI/SNF-directed probes PFI-3 and I-BRD9 — were compared between ARID1A-mutant and wild-type cell lines by Mann–Whitney test, separately in lung-only and pan-cancer strata. The lung stratum is a subset of the pan-cancer stratum for all thirteen compounds, so the two sets of tests are not independent and Benjamini–Hochberg correction was applied within each stratum rather than across the pooled 26 comparisons. Minimum detectable differences were computed post hoc at $\alpha = 0.05$ and 80% power over a plausible range of within-group standard deviations (0.06–0.10 AUC). To establish whether the assay had dynamic range, each compound's mean AUC was ranked against all 286 compounds profiled in the same lung panel.

Genotype-conditional dependency. For each BAF subunit, DepMap gene effect was compared between ARID1A-mutant and wild-type lines within each tissue having at least six mutant lines (five qualified: endometrium, ovary, large intestine, stomach, lung), by Mann–Whitney with global Benjamini–Hochberg correction. Subunits are summarised by the number of tissues in which mutant lines are the more dependent group, a categorical read-out distinct from the continuous gradient used elsewhere.

Virtual knockout and Mendelian randomization. Geneformer [46] on GSE131907 malignant epithelial cells, run on a 2,000-cell downsample and on all 7,270 cells; scGPT [47] on the same cells, with transcriptional-neighbour percentiles the rank of each SWI/SNF subunit's cosine similarity to ARID1A among all 13,032 genes in the embedding. Two-sample Mendelian randomization with MR-Base [49], Wald ratio.

Author contributions

Zhiyong Tao: Conceptualization; Methodology; Software; Formal analysis; Investigation; Visualization; Writing — original draft. **Juan Xiong:** Methodology; Formal analysis; Validation; Data curation; Investigation; Writing — original draft. **Yiwei Zhang:** Methodology; Software; Formal analysis; Validation; Data curation; Visualization; Writing — original draft. **Chulin Zhong:** Data curation; Formal analysis; Validation; Visualization. **Wenkai Gao:** Data curation; Investigation;

Validation; Writing — review & editing. **Guohua Fan:** Conceptualization; Supervision; Project administration; Resources; Writing — review & editing.

Zhiyong Tao, Juan Xiong and Yiwei Zhang contributed equally to this work. All authors read and approved the final manuscript.

Key Points

- ARID1A-contacting subunits show higher co-dependency than non-contacting subunits, but contact count does not predict co-dependency within resolved contacts; structure is a binary prior, not a quantitative rule.
- Curated protein-complex module definitions are sub-optimal for dependency analysis; refining the SWI/SNF core raises its correlation from 0.475 to 0.584 while dropping two of eight genes, and the refinement generalises to PRC2.
- Module-score co-dependency is inflated unless the anchor is excluded from its own module, which reversed conclusions in two of the modules tested here.
- Five methodologically distinct analyses with pan-essentiality and lineage controls assign 14/14 anchors of known membership to their correct mutually exclusive module across two complexes.
- Genotype-conditional tests do not survive multiple-testing correction, so the study maps co-essential organization rather than synthetic lethality.
- Transcript-level data cannot resolve complex specificity, and AlphaFold-Multimer's ipTM does not track interface accuracy within a large assembly.

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Conflict of interest

The authors declare no competing interests.

Declaration of generative AI use

During the preparation of this work the authors used generative AI assistants (Anthropic Claude and OpenAI Codex) to write and execute analysis code, to generate figures from the resulting data, and to draft and revise manuscript text. All analyses were specified by the authors, all outputs were checked against the underlying data, and the authors reviewed and edited the content and take full responsibility for it. Generative AI was not used to produce or alter any primary data, and is not listed as an author. The machine-learning methods described in Methods (Geneformer, scGPT, AlphaFold-Multimer, gradient boosting with SHAP) are research tools applied to the data and are distinct from the assistance declared here.

Data and code availability

This study analysed only publicly available, previously published human datasets under the accessions listed in Methods; no new human or animal experiments were performed and no new ethical approval was required. Versioned analysis code, a pinned software environment, random seeds, automated numerical checks, compact derived tables and the submitted manuscript snapshot are publicly available at `https://github.com/ArleneHarrison/arid1a-baf-collateral-dependency` (release v1.0.4) and archived at Zenodo (`https://doi.org/10.5281/zenodo.22218065`). Raw source datasets are not redistributed and remain available from the persistent accessions and URLs listed in the repository data manifest. No claim in the manuscript depends on data available only on request.

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